Aditya Rao

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Education

McMaster University Sep 2023 – Apr 2028 (Expected)
B.Eng in Software Engineering (CO-OP) *Hamilton, ON*

- **3.99/4.00 CGPA** - Dean's Honour List
- **Relevant Coursework:** Data Structures and Algorithms, Concurrent Systems Design, Software Requirements, Large System Design, Computer Architecture, Software Testing, Databases

Experience

IBM May 2025 – Aug 2026
Software Developer Intern, WebSphere Application Server *Markham, ON*

- Rearchitected legacy software-patch publishing into containerized Java services with Kafka workers and PostgreSQL, reducing full-batch publishing from half a day to around 10 minutes.
- Built an AI agent to investigate build failures using logs, traces, and source code, invoke team-specific diagnostic workflows, and recommend fixes, saving ∼4 hours per failure.
- Rebuilt customer-issue tracking to address synchronization bugs, keeping the upstream support system authoritative for ticket data and storing approval requests separately.
- Added versionless feature installation to a Java application server's installation utility and rewrote its installation functional test suite in JUnit for easier maintenance and debugging.
- Built a centralized work-allocation platform with developer submissions, manager approvals, and reporting exports, replacing fragmented spreadsheets across the organization and saving ∼60 minutes per person weekly.
- Delivered 27 merged open-source PRs to OpenLiberty and 100+ merged internal PRs across WebSphere products and developer infrastructure.

McMaster University Sep 2026 – Dec 2026
Teaching Assistant - Digital Systems *Hamilton, ON*

- Led digital systems labs for 50+ students, graded prelabs and demonstrations, and supported Verilog design and FPGA debugging.
- Validated and improved lab guides covering Nios II systems and embedded C programs for DRAM read/write testing on DE1-SoC FPGAs.

Medium AI May 2024 – Aug 2024
Full Stack Developer Intern *Hamilton, ON*

- Engineered real-time doctor-patient voice-based transcription with **gRPC** and **Firebase**, achieving **sub-500ms streaming latency** by load balancing **OpenAI Whisper**.
- Implemented FFmpeg audio preprocessing and Firebase transcript synchronization to stream transcription updates to the web UI.

Projects

BYoAI Deployment Platform | | *Go, TypeScript, Docker, MCP, OpenTelemetry*

- Built an open-source platform deploying AI agents in sandboxed **Docker** environments, using a **Go** daemon (**40 MB RSS**) for isolated execution and a **TypeScript/Bun** service for model orchestration and tool routing.
- Engineered pluggable **event routing** across Redis, RabbitMQ, and Kafka, and a custom **MCP-over-RPC transport** tunneling tool servers inside remote Docker containers.
- Built a React 19 web UI and Go terminal TUI with OAuth2 (confidential + PKCE), declarative tool ACLs, and human-in-the-loop approval gates.

WebCraft | | *C++, CMake, Vcpkg, Asynchronous Programming, Github Actions*

- Authored an asynchronous runtime in **C++23** featuring asynchronous timers, task scheduling, and high-performance file and socket (TCP and UDP) I/O modules.
- Sustained **160K req/s** on synthetic **wrk** HTTP benchmarks via an **io_uring** event loop on Linux.
- Automated CI using **gtest (130+ tests)**, **GitHub Actions CI/CD** (Linux, Windows, macOS).

Technical Involvement

Sys-Admin and Fullstack Developer Executive Nov 2024 – Present
IEEE McMaster Student Branch *Hamilton, ON*

- Set up an **RPi 3B+ cluster** handling up to 100 concurrent clients with **<50ms response latency**.
- Provisioned **CI/CD pipelines** to build and deploy container images to a **Kubernetes cluster**.
- Developed the IEEE branch website using Next.js with GitHub Projects/Issues as an admin CMS.

Technical Skills

Languages: Java, JavaScript/TypeScript, Python, C#, C++, Golang, SQL, C, Verilog
Frameworks: Spring Boot, Jakarta EE, MicroProfile, Quarkus, Hibernate, Express, Next.js, FastAPI, Flask
Developer Tools: AWS, GCP, Git, Docker, Kafka, OpenShift, Kubernetes, Redis, PostgreSQL, Infinispan